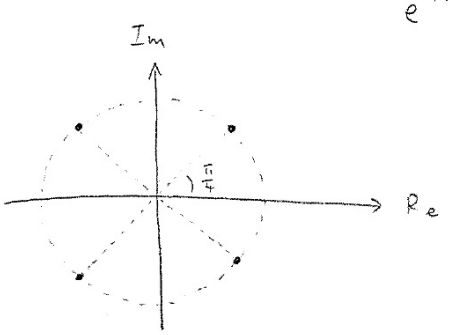
1. $z = re^{i\theta} \rightarrow z^4 + 81 = r^4 e^{4i\theta} + 3^4 = 0$

$r = 3$

$e^{4i\theta} = -1 \rightarrow 4\theta = (2n-1)\pi \quad n = 1, 2, 3, \cdots$

$\theta = \frac{2n-1}{4}\pi$

$= \frac{1}{4}\pi, \frac{3}{4}\pi, \frac{5}{4}\pi, \frac{7}{4}\pi$



$-81 = 3^4 e^{i\pi + i2\pi k}$

2. $\text{左边} = \prod_{k=1}^{m-1} \sin\frac{k\pi}{m} = \prod_{k=1}^{m-1} \mathrm{Im}(w^k)$ $\qquad w = e^{i\frac{\pi}{m}}$

$$= \prod_{k=1}^{m-1} \frac{w^k - \bar{w}^k}{2i} = \frac{1}{(2i)^{m-1}} \prod_{k=1}^{m-1} (w^k - \bar{w}^k) = \frac{1}{(2i)^{m-1}} \prod_{k=1}^{m-1} (w^k - w^{2m-k})$$

$$= \frac{e^{-i\frac{m-1}{2}\pi}}{2^{m-1}} w^{\frac{1}{2}m(m-1)} \prod_{k=1}^{m-1} (1 - u^{m-k}) \qquad u = w^2 = e^{i\frac{2\pi}{m}}$$

$$= \frac{1}{2^{m-1}} \prod_{k=1}^{m-1} (1 - u^k)$$

$$= \frac{1}{2^{m-1}} \cdot \frac{x^m - 1}{x - 1}\Big|_{x=1}$$

$$= \frac{1}{2^{m-1}} \cdot (x^{m-1} + x^{m-2} + \cdots + 1)\Big|_{x=1}$$

$$= \frac{m}{2^{m-1}} \quad \text{J}$$

3. (a) $z = e^{\omega} = re^{i\theta}$, $\omega = u + iv \ (= \ln z)$

$$e^{\omega} = e^{u} \cdot e^{iv} \rightarrow u = \ln r \ , \ v = \theta + 2n\pi$$

i.e. $z = re^{i\theta} \rightarrow \ln z = \ln r + i(\theta + 2\pi n)$

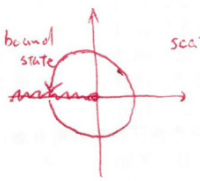
(b) $\ln(1-i) = \ln\sqrt{2} + i\left(-\frac{\pi}{4} + 2\pi n\right)$

主値 $= \ln\sqrt{2} - \frac{\pi}{4} i$

$\arg z = [0, 2\pi]$

$= (-\pi, \pi)$

$(-\pi, \pi]$



$\mathbb{C}^{\times} = \mathbb{C} -$

$z^{\alpha} := e^{\alpha \ln z}$

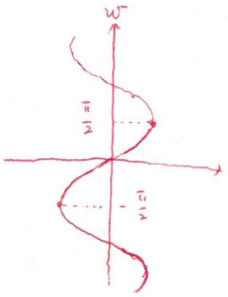
4. (a) $z = \sin w = \dfrac{e^{iw} - e^{-iw}}{2i}$

$u^2 - 2iz u - 1 = 0$

$\rightarrow u = iz \pm \sqrt{1 - z^2}$

$\sin^{-1} 0 = 0$ & plus

$z = \tanh w = \dfrac{e^{iw} + e^{-iw}}{e^{iw} \mp e^{-iw}}$



(b) $i^i = (e^{+i\frac{\pi}{2}})^i = e^{-\frac{\pi}{2}}$

$= e^{i \ln i}$

$= e^{i\{i\frac{\pi}{2} + i2\pi k\}}$

$= e^{-\frac{\pi}{2}} (e^{-2\pi})^k$



5. (a) $z = (1+i)^i = \left\{\sqrt{2}\, e^{i(\frac{\pi}{2}+2n\pi)}\right\}^i = \sqrt{2}^i e^{-(2n+\frac{1}{2})\pi}$

$= e^{-(2n+\frac{1}{2})\pi} e^{i\ln\sqrt{2}}$

$0^i = ?$

(b) $z = 1^{\sqrt{2}} = (e^{2\pi n})^{\sqrt{2}} = e^{2\sqrt{2}n\pi}$

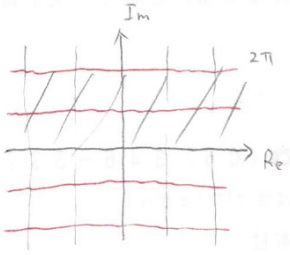
(c) $z = (1-i)^{1+i} = \left\{\sqrt{2}\, e^{i(-\frac{\pi}{2}+2n\pi)}\right\}^{1+i} = \sqrt{2}^{1+i}\, e^{-(2n-\frac{1}{2})\pi}$

$= \sqrt{2}\, e^{-(2n-\frac{1}{2})\pi} e^{i\ln\sqrt{2}}$

$|0^i| = (0^i)\cdot(0^i)^* = 0^{i-i} = 0^0$ 不定

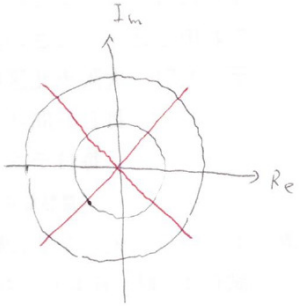
6.

a)

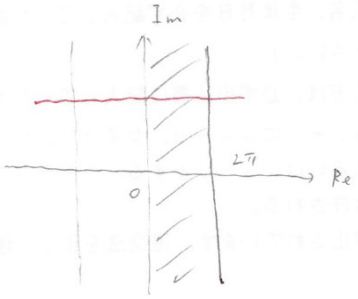


a)

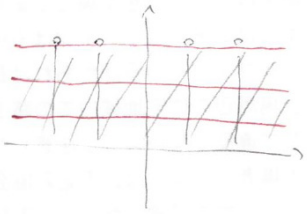
c)



b)



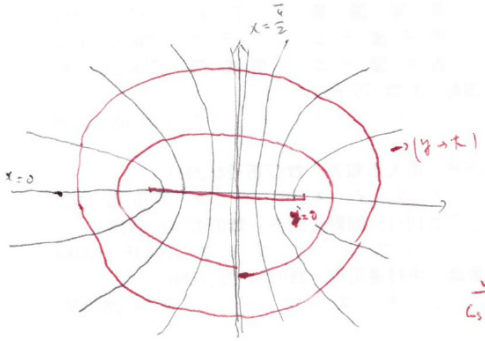
c)





$$\cos z = \frac{e^{iz}+e^{-iz}}{2} = \frac{e^{ix}e^{-y}+e^{-ix}e^{y}}{2}$$

$$= \cosh y \cos x + i \sinh y \sin x$$



直交座標変換

解析写像

$$\frac{u^2}{\cosh y^2} + \frac{v^2}{\sinh y^2} = 1$$

$$\frac{u^2}{\cos^2 x} - \frac{v^2}{\sin^2 x} = 1$$

7.

$$df = \frac{\partial f}{\partial z}dz + \frac{\partial f}{\partial \bar{z}}d\bar{z} \qquad \begin{cases} dx = \frac{1}{2}(dz + d\bar{z}) \\ dy = \frac{1}{2i}(dz - d\bar{z}) \end{cases}$$

$$= \frac{1}{2}\left(\frac{\partial f}{\partial x} + \frac{1}{i}\frac{\partial f}{\partial y}\right)dz + \frac{1}{2}\left(\frac{\partial f}{\partial x} - \frac{1}{i}\frac{\partial f}{\partial y}\right)d\bar{z}$$

(a) $z = x + iy \quad (x, y \in \mathbb{R})$

$$\begin{cases} \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \\ \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y} \end{cases} \quad \cdots (*)$$

$\rightarrow \Delta u = \Delta v = 0$

$\left(\Delta = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right)$

(b) $\frac{\partial}{\partial \bar{z}} = \frac{\partial x}{\partial \bar{z}}\frac{\partial}{\partial x} + \frac{\partial y}{\partial \bar{z}}\frac{\partial}{\partial y} = \frac{1}{2}\left(\frac{\partial}{\partial x} - \frac{1}{i}\frac{\partial}{\partial y}\right)$

$\cdots\rightarrow \frac{\partial f}{\partial \bar{z}} = 0 \iff (*)$

$w = f(z)$
複素数の関数

$u + iv = f(x + iy) = u(x, y) + iv(x, y)$
$\neq f(x, y)$

解析微分？

$$f'(z) = \lim_{h \to 0} \frac{f(z+h) - f(z)}{h} = \lim_{|\Delta x|^2 + |\Delta y|^2 \to 0} \frac{u(x+\Delta x, y+\Delta y) + iv(x+\Delta x, y+\Delta y) - u(x,y) - iv(x,y)}{\Delta x + i\Delta y}$$

$(x+\Delta x) + i(y+\Delta y)$ ← $x + iy$

$\Delta y = 0$ とする

$$= \lim_{\Delta x \to 0} \frac{u(x+\Delta x, y) + iv(x+\Delta x, y) - u(x,y) - iv(x,y)}{\Delta x}$$

$$= \frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x}$$

方向指定なら $\Delta x + i\Delta y = \rho e^{i\theta}$

8. (a) $dv = \frac{\partial v}{\partial x}dx + \frac{\partial v}{\partial y}dy$

$= -\frac{\partial u}{\partial y}dx + \frac{\partial u}{\partial x}dy$ (∵ Cauchy-Riemann condition)

$= 2x\,dx + 2(1-y)\,dy$ $\quad u = 2x(1-y)$

$\therefore v(x,y) = x^2 - (y-1)^2 + \text{const}$

$f = u + iv = 2x(1-y) + i(x^2 - y^2 + 2y) + \text{const}$

$= 2(x+iy) + i(x^2 - y^2) - 2xy + \text{const}$

$= 2z + z^2 + \text{const}$

(b) $\psi = \ln(x^2+y^2)$ 는 조화함수.

$\frac{\partial^2 \psi}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{2x}{x^2+y^2}\right) = 2\frac{-x^2+y^2}{(x^2+y^2)^2}$ 또) $\Delta\psi = 0$ ⌟

(harmonic function)

$d\phi = \frac{\partial \phi}{\partial x}dx + \frac{\partial \phi}{\partial y}dy$

$= -\frac{\partial \psi}{\partial y}dx + \frac{\partial \psi}{\partial x}dy$

$= \frac{-2y}{x^2+y^2}dx + \frac{2x}{x^2+y^2}dy$

$\phi = 2\left\{\tan^{-1}\left(\frac{y}{x}\right) - \tan^{-1}\left(\frac{x}{y}\right)\right\} + \text{const}$ $\quad \because \int \frac{dt}{t^2+1} = \tan^{-1}t$

$z = x+iy$, $a = a+ib$ $\quad = 2\tan^{-1}\left(\frac{y}{x}\right) + \text{const}$

$\psi = \ln\{(x-a)+i(y-b)\}\{(x-a)-i(y-b)\}$

$= \ln\{(z-\alpha)(\bar{z}-\bar{\alpha})\} = \ln|z-\alpha|^2$

$= \ln(z-\alpha) + \ln(\bar{z}-\bar{\alpha})$

$\phi = -i\ln(\bar{z}-\bar{\alpha}) + (\text{c.c})$

$= -i\ln(\bar{z}-\bar{z}) + i\ln(z-\alpha)$

$= i\ln\frac{z-\alpha}{\bar{z}-\bar{\alpha}}$

$f = \phi + i\psi = 2i\ln(z-\alpha)$ ⌟

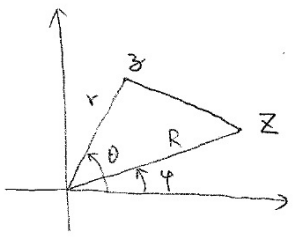
$t = \tan\theta \quad \theta = \tan^{-1}t$

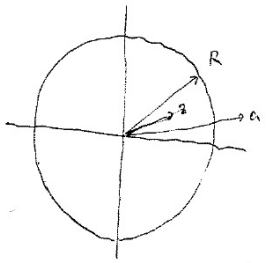
$dt = (1+\tan^2\theta)\,d\theta$

$d\theta = \frac{dt}{1+t^2}$ $\quad \frac{d\theta}{dx} = \frac{d}{dt}\tan^{-1}t$

$d\left(\tan^{-1}\left(\frac{x}{y}\right)\right) = \frac{1}{1+\left(\frac{x}{y}\right)^2}\cdot\frac{-x\,dy + y\,dx}{y^2}$

9.





$$\psi = \frac{R^2 - r^2}{R^2 + r^2 - 2Rr\cos(\theta - \varphi)} = \frac{|Z|^2 - |z|^2}{|Z - z|^2} = \frac{Z\bar{Z} - z\bar{z}}{(Z - z)(\bar{Z} - \bar{z})}$$

$$\frac{\partial^2 \psi}{\partial \bar{z} \partial z} = \frac{\partial}{\partial \bar{z}} \left(\frac{Z}{(z - Z)^2} \right) = 0 \quad \text{Ans)}$$

$$= \frac{1}{2} \left(\frac{z + z}{Z - z} + \frac{\bar{Z} + \bar{z}}{\bar{Z} - \bar{z}} \right)$$

$$= \mathrm{Re} \frac{z + Z}{z - Z}$$

$$\Delta \psi = 2 \left(\frac{\partial^2}{\partial z \partial \bar{z}} + \frac{\partial^2}{\partial \bar{z} \partial z} \right) \psi = 0 \quad (\text{q.e.d})$$

$$df = \frac{\partial f}{\partial z} dz + \frac{\partial f}{\partial \bar{z}} d\bar{z}$$

$$\begin{cases} dz = dx + i dy \\ d\bar{z} = dx - i dy \end{cases}$$

$$= \left(\frac{\partial}{\partial z} + \frac{\partial}{\partial \bar{z}} \right) f \, dx + i \left(\frac{\partial}{\partial z} - \frac{\partial}{\partial \bar{z}} \right) f \, dy$$

$$\rightarrow \quad \frac{\partial}{\partial x} = \frac{\partial}{\partial z} + \frac{\partial}{\partial \bar{z}} \quad , \quad \frac{\partial}{\partial y} = i \left(\frac{\partial}{\partial z} - \frac{\partial}{\partial \bar{z}} \right)$$

10. f の Laurent 展開 $f(z) = \sum_{-\infty}^{\infty} a_n (z-a)^n \quad 0 < |z-a| < R$

一様収束 → 項別積分可能

$$a_{-1} = \operatorname*{Res}_{z=a} f(z) = \mathrm{Res}(f, a) \qquad \text{Residue (留数)}$$

a が f の n 次の極 → $\mathrm{Res}(f,a) = \frac{1}{(n-1)!} \lim_{z\to a} \frac{d^{n-1}}{dz^{n-1}} \left[(z-a)^n f(z)\right]$

$f(z) = \sum_{-n}^{\infty} a_k (z-a)^k$ のとき a は f の n 次の極

$$\to \lim_{z\to a} (z-a)^m f(z) = \begin{cases} 0 & (m > n) \\ \text{not } 0 & (m = n) \\ \pm\infty & (m < n) \end{cases}$$

1次の極 $\lim_{z\to a} (z-a)^2 f(z) = 0$

$\lim_{z\to a} (z-a) f(z) = \mathrm{Res}(f, a) \neq 0$

(a) $f(z) = z(z-2)(z+1)^{-2}(z-2i)^{-1}(z+2i)^{-1}$

$a = \pm 2i$

$$\mathrm{Res}(f, 2i) = \left.\frac{z(z-2)}{(z+1)^2(z+2i)}\right|_{z=2i} = \frac{7+i}{25}$$

$$\mathrm{Res}(f, -2i) = \left.\frac{z(z-2)}{(z+1)^2(z-2i)}\right|_{z=-2i} = \frac{7-i}{25}$$

$$\mathrm{Res}(f, -1) = -\frac{14}{25}$$

10. n次の零点 : $f^{(k)}(a)=0 \quad (0\le k\le n-1) \wedge f^{(n)}(a)\neq 0$

aが $f(z)=\dfrac{P(z)}{Q(z)}$ の1次の極で $P(a)\neq 0$ ならば

$$Res(f,a)=\frac{P(a)}{Q'(a)}$$

(証明)

$$Res(f,a)=\lim_{z\to a}\frac{z-a}{Q(z)-\underbrace{Q(a)}_{\to 0}}P(a)$$

(b) $f(z)=\dfrac{1}{\cosh(z)}$ $\quad \cosh(z)=0 \to a=i\left(\frac{\pi}{2}+\pi k\right) \quad (e^a=(-1)^k i)$

$\cosh'(a)=\sinh(a)=(-1)^k i$

$$Res\left(f, i\pi\left(\tfrac{1}{2}+k\right)\right)=\frac{1}{(-1)^k i}=-(-1)^k i$$

(c) $f(z)=\dfrac{\cos(z)}{\sin(z)}$ $\quad \sin(z)=0 \to a=i\pi k \quad (e^a=(-1)^k)$

$\sin'(a)=\cos(a)=(-1)^k$

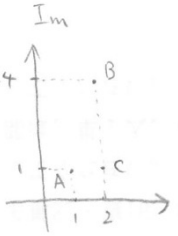
$$Res(f, i\pi k)=1$$

(d) $f(z)=\dfrac{e^z}{\sin^2 z}$ $\quad \sin^2 z$ の零点は $f(z)$ の2次のpole

$$Res(f, i\pi k)\neq\frac{e^{i\pi k}}{\cos(i\pi k)}=*$$

$$\lim_{z\to n\pi}\frac{d}{dz}(z-n\pi)^2\frac{e^z}{(\sin z)^2}=\lim_{x\to 0}e^{n\pi}\frac{d}{dx}\cdot\frac{x^2e^x}{(\sin x)^2}=e^{n\pi}\lim_{x\to 0}\left\{\frac{d}{dx}\left(\frac{x}{\sin x}\right)^2\cdot e^x+\underbrace{\left(\frac{x}{\sin x}\right)^2}_{\to 1}e^x\right\}=e^{n\pi}$$

11. $I = \int_A^B z^2\,dz$



経路によらない → 原始関数の存在

$f(z) = u + iv$ $\quad z = x + iy$

$\int f(z)\,dz = \int (u+iv)\,d(x+iy)$

$= \int (u\,dx - v\,dy) + i(v\,dx + u\,dy)$

外微分形式の理論

$w = P(x,y)\,dx + Q(x,y)\,dy$: 微分形式

$w = f(z)\,dz = (u+iv)\,d(x+iy)$

$\to (u+iv) \wedge d(x+iy)$

$dw = d(u+iv) \wedge d(x+iy) + \underbrace{(u+iv) \wedge d^2(x+iy)}_{=0}$

$= (du + i\,dv) \wedge (dx + i\,dy)$

$= (du \wedge dx - dv \wedge dy) + i(du \wedge dx + du \wedge dy)$

$= \left(\frac{\partial u}{\partial x}dx + \frac{\partial u}{\partial y}dy\right) \wedge dx = \frac{\partial u}{\partial y}dy \wedge dx$

$= \frac{\partial u}{\partial y}dx \wedge dy - \frac{\partial v}{\partial x}dx \wedge dy - i\frac{\partial v}{\partial y}dx \wedge dy + i\frac{\partial u}{\partial x}dx \wedge dy$

$= \left[-\left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}\right) + i\left(\frac{\partial u}{\partial x} - \frac{\partial v}{\partial y}\right)\right] dx \wedge dy$

$= 0$: f C.R. cond. is satisfied

$\Updownarrow$

$dw = 0$: 外微分形式が閉じている

Stokes の定理

$\int_{C=\partial R} w = \int_R dw$

領域 D : 連結された開集合

単連結領域 : D

D内にあるすべての閉曲線 C の内部の点が D に属する

(すべての閉じた経路が1点にホモトープである)

$I = \int_A^B z^2\,dz = \left[\frac{1}{3}z^3\right]_A^B$

D : 単連結領域で $f(z)$ が解析的ならば

$\forall C$: 閉曲線 に対して

$\int_C f(z)\,dz = 0$

(a) $y = x^2 \to z = t + it^2$ $\quad dz = (1 + 2it)\,dt$

$$I = \int_{t=1}^{t=2} t^2(1+it)^2(1+2it)\,dt$$

$dx \wedge dy = -dy \wedge dx$

$$= \int_{t=1}^{t=2} \left\{t^2(1-5t^2) + 2i\,t^3(2-t^2)\right\} dt$$

$$= -\frac{86}{3} - 6i$$

(b) $\overline{AB} \to z = t + 3it$ $\quad dz = (1+3i)\,dt$

$$I = \int_{t=1}^{t=2} t^2(1+3i)^3\,dt = -\frac{182}{3} - 42i$$

(c) $\overline{AC}$ and $\overline{CB} \to \begin{cases} z = t + i & dz = dt \\ z = 2 + it & dz = i\,dt \end{cases}$

$$I = \int_{t=1}^{t=2} (t+i)^2\,dt + \int_{t=1}^{t=2} (2+it)^2\,i\,dt$$

$$= \left(\frac{4}{3} + 3i\right) + \left(-6 + \frac{5}{3}i\right) = -\frac{14}{3} + \frac{14}{3}i$$

(d) $z = u + iv$ $\quad dz = du + i\,dv$

$$I = \int \left\{3(u^2 - v^2) + 6iuv\right\}(du + i\,dv)$$

$$= \left\{ \begin{array}{l} \int_1^4 (7 - 3v^2 + 9iv)\,dv \\ \int_1^2 (9u^2 - 63 + 45iu)\,du \end{array} \right\} = -42 + \frac{135}{2}i$$

12. $f(z) = \frac{1}{iz} R\left[\frac{1}{2}\left(z+\frac{1}{z}\right), \frac{1}{2i}\left(z-\frac{1}{z}\right)\right]$ が $C: z = e^{it}$ 上に極を持たないとき．C内の相異なる極の全体を α_j とすると

$$\int_0^{2\pi} R(\cos t, \sin t)\,dt = \int_C f(z)\,dz \qquad \left(\frac{dz}{iz} = dt\right)$$

$$= 2\pi i \sum_j \mathrm{Res}(f, \alpha_j)$$

(a) $$f(z) = \frac{1}{iz} \cdot \frac{1}{\left(a + \frac{b}{2}\left(z+\frac{1}{z}\right)\right)^3} = \frac{-8iz^2}{(bz^2+2az+b)^3} = \frac{-8iz^2}{b^3(z-\alpha_+)^3(z-\alpha_-)^3}$$

3次の極

$$\alpha_\pm = \frac{-a \pm \sqrt{a^2-b^2}}{b} \qquad (a > |b|) \qquad \underset{b>0}{\alpha_- < \alpha_+ < 0}, \quad \alpha_+ \cdot \alpha_- = 1$$

$$\to |\alpha_+| < 1 < |\alpha_-|$$

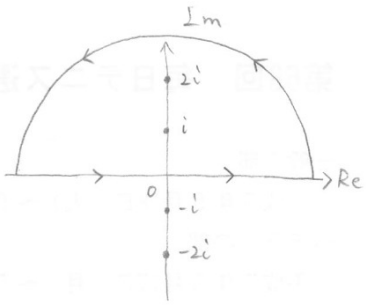
$$\mathrm{Res}(f, \alpha_+) = \frac{1}{2!} \lim_{z\to\alpha_+} \frac{d^2}{dz^2}(z-\alpha_+)^3 f(z)$$

$$= \frac{1}{2} \lim_{z\to\alpha_+} \frac{-16i(z^2+4\alpha_- z+\alpha_-^2)}{b^3(z-\alpha_-)^5} = \frac{-i(2a^2+b^2)}{2(a^2-b^2)^{5/2}}$$

$$\therefore \int_0^{2\pi} \frac{d\theta}{(a+b\cos\theta)^3} = +2\pi i \cdot \frac{2a^2+b^2}{2(a^2-b^2)^{5/2}}$$

$$\| \qquad \int_0^{\pi} \frac{d\theta}{(a+b\cos\theta)^3} + \int_0^{\pi} \frac{d\theta}{(a-b\cos\theta)^3}$$

12 (b)

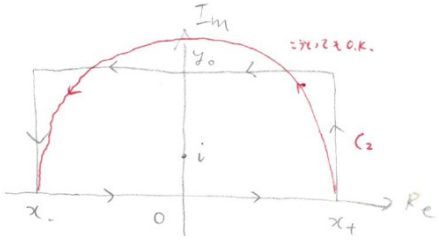


$$\frac{1}{(z^2+1)^2(z^2+4)}$$

$$= -\frac{i}{36(z-i)} - \frac{i}{36(z-2i)} - \frac{1}{12(z-i)^2} + \frac{i}{36(z+i)} - \frac{1}{12(z+i)^2} + \frac{i}{36(z+2i)}$$

$$\int_{-\infty}^{\infty} \frac{dx}{(x^2+1)^2(x^2+4)} = \int_C \frac{dz}{(z^2+1)^2(z^2+4)} = 2\pi i \left\{ -\frac{i}{36} - \frac{i}{36} \right\} = \frac{\pi}{9}$$

(c) $$\int_{-\infty}^{\infty} \frac{\cos x}{x^2+1}\,dx = \int_\gamma \frac{e^{iz}}{z^2+1}\,dz$$



$= 2\pi i \operatorname{Res}\left(\frac{e^{iz}}{z^2+1}, i\right)$

$= 2\pi i \cdot \frac{e^{-1}}{2i} = \frac{\pi}{e}$

i は1次の極

$\int_{-\infty}^{\infty} f(z)e^{iz}\,dz$

$\max|f(re^{i\theta})| = M(r)$ ← $f(z)$ が上半平面で解析的, $\lim_{|z|\to\infty} f(z) = 0$

$\lim_{r\to\infty} M(r) = 0$

$z = x_+ + iy$, $z = x + iy_0$, $z = x_- + iy$

$$\left|\int_{\gamma'} \frac{e^{iz}}{z^2+1}\,dz\right| \le \left|\int_0^{y_0} \frac{M}{|z|} e^{-y}\,dy\right| + \left|\int_{x_+}^{x_-} \frac{M}{|z|} e^{-y_0}\,dx\right| + \left|\int_{y_0}^{0} \frac{M}{|z|} e^{-y}\,dy\right|$$

$|S_{C_2}| \le \int_{-\infty}^{\infty} |f(Re^{i\theta})| e^{-R\lambda\theta} R\,d\theta$

$\le M(R) \boxed{R\int_0^{\pi} e^{-R\lambda\theta}\,d\theta}$

$\le \frac{M}{x_+} + \frac{M}{y_0}(x_+ - x_-) + \frac{M}{x_-} \to 0$

$= 2R\int_0^{\frac{\pi}{2}} e^{-R\lambda\theta}\,d\theta \le 2R\int_0^{\frac{\pi}{2}} e^{-\frac{2R}{\pi}\theta}\,d\theta = 2R\left[\frac{e^{-\frac{2R}{\pi}\theta}}{-\frac{2R}{\pi}}\right]_0^{\frac{\pi}{2}} = \pi(1-e^{-R}) < \pi$

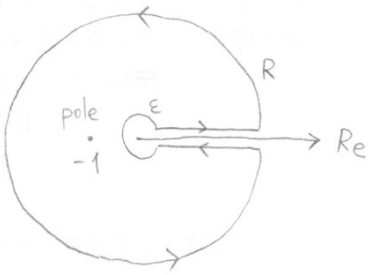
☺ $\left|\frac{1}{z^2+1}\right| \le \frac{M}{|z|}$

12 (d)

$$\int_\gamma \frac{(\ln z)^2}{(1+z)^3}\,dz = 2\pi i\,\mathrm{Res}\left(\frac{(\ln z)^2}{(1+z)^3}, -1\right)$$

$$= 2\pi i \lim_{z\to -1} \frac{1}{2!}\frac{d^2}{dz^2}(\ln z)^2$$

$$= i\pi \lim_{z\to -1} \frac{2}{z^2}(1-\ln z) = 2\pi i + 2\pi^2$$



$$\left|\int_{\gamma_R} \frac{(\ln z)^2}{(1+z)^3}\,dz\right| \le \frac{M}{R^3}\sqrt{(\ln R)^2+4\pi^2}\cdot 2\pi R \longrightarrow 0 \quad (R\to\infty)$$

$$\left|\int_{\gamma_\epsilon} \frac{(\ln z)^2}{(1+z)^3}\,dz\right| \le M\sqrt{(\ln \epsilon)^2+4\pi^2}\cdot 2\pi\epsilon \longrightarrow 0 \quad (\epsilon\to 0)$$

$$\int_{\gamma-(\gamma_R+\gamma_\epsilon)} \frac{(\ln z)^2}{(1+z)^3}\,dz = \int_0^\infty \frac{(\ln x)^2}{(1+x)^3}\,dx + \int_\infty^0 \frac{(\ln x+2\pi i)^2}{(1+x)^3}\,dx$$

$$= -4\pi i \underbrace{\int_0^\infty \frac{\ln x}{(1+x)^3}\,dx}_{=-\frac{1}{2}} + 4\pi^2 \underbrace{\int_0^\infty \frac{dx}{(1+x)^3}}_{\frac{1}{2}}$$

$\int R(z)(\ln z)^n dz$

C と c の寄与が消える条件

$\lim_{|z|\to\infty} zR(z) = 0$

$\frac{dz}{z} = i\,d\theta$

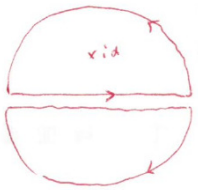
$R(z) \to \frac{1}{z^n}$ $(|z|\to\infty)$

$\lim_{|z|\to\infty} zR(z)(\ln z)^n = \lim_{R\to\infty} R\cdot\frac{1}{R^n}(\ln R)^n \to 0$

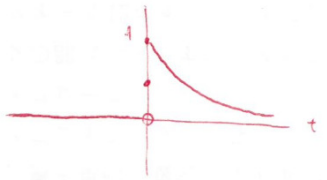
$\lim_{\varepsilon\to 0} R(\varepsilon)\,\varepsilon(-\ln\varepsilon)^n = R(0)\lim_{x\to\infty} e^{-x}x^n = 0$

14. (a) $\lim_{|z|\to\infty} f(z) = 0 \iff \max_{z=Re^{i\theta}} |f(z)| = M(R) \to 0$

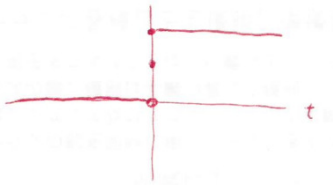
(b)



$$\theta_\alpha(t) = \frac{1}{2\pi i}\int_{-\infty}^{\infty} \frac{e^{its}}{s-i\alpha}\,ds = e^{it(i\alpha)} = \begin{cases} e^{-\alpha t} & \text{for } t>0 \\ 0 & t<0 \end{cases}$$



$\downarrow (\alpha \to 0)$



$$\theta'(t) = \delta(t)$$

$$\theta_\alpha(0) = \frac{1}{2\pi i}\int_{-\infty}^{\infty}\frac{ds}{s-i\alpha} = \frac{1}{2\pi i}\int_{-\infty}^{\infty}\frac{s+i\alpha}{s^2+\alpha^2}\,ds = \frac{\alpha}{2\pi}\int_{-\infty}^{\infty}\frac{ds}{s^2+\alpha^2} = \frac{1}{2}$$

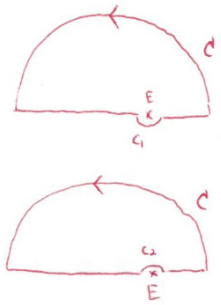
$$\int_{-\infty}^{\infty}\theta'(x)f(x)\,dx = \int_{-\infty}^{\infty}\left(\frac{d}{dx}\theta(x)\right)f(x)\,dx = \int_{-\infty}^{\infty}\left[\frac{d}{dx}\{\theta(x)f(x)\} - \theta(x)\frac{df(x)}{dx}\right]dx$$

$$= \theta(x)f(x)\Big|_{-\infty}^{\infty} - \int_{-\infty}^{\infty}\theta(x)\frac{df(x)}{dx}\,dx$$

$$= f(\infty) - \int_{-\infty}^{\infty}\frac{df(x)}{dx}\,dx = f(\infty) - [f(x)]_0^{\infty} = f(0) = \int_{-\infty}^{\infty}\delta(x)f(x)\,dx$$

$$\Rightarrow \quad \theta'(x) = \delta(x)$$

15.



$$\oint \frac{f(E')}{E'-E}\,dE'$$

$$\int_C + \int_{-\infty}^{E-\delta} + \int_{E+\delta}^{\infty} + \int_{C_1} = 2\pi i\, f(E) = \int_{C_+} = \int_{-\infty}^{\infty} dE' \frac{f(E')}{E'-(E+i\varepsilon)}$$

$$\int_C + \int_{-\infty}^{E-\delta} + \int_{E+\delta}^{\infty} + \int_{C_2} = 0 = f_{C_-} = \int_{-\infty}^{\infty} dE' \frac{f(E')}{E'-(E-i\varepsilon)}$$

$$P\int_{-\infty}^{\infty} \frac{f(E')}{E'-E}\,dE' = \lim_{\delta\to 0}\left\{\int_{-\infty}^{E-\delta} \frac{f(E')}{E'-E}\,dE' + \int_{E+\delta}^{\infty} \frac{f(E')}{E'-E}\,dE'\right\} \qquad \frac{1}{2}\left\{\int_{C_+} + \int_{C_-}\right\} = P\int = \pi i f(E)$$

$$\frac{1}{2}\left\{\int_{C_+} - \int_{C_-}\right\} = \pi i f(E)$$

$$\int_{C_1} + \int_{C_2} = i f(E)\left\{\int_{-\pi}^{0} d\theta + \int_{\pi}^{0} d\theta\right\} = 0$$

$$\int_{C_1} - \int_{C_2} = 2\pi i\, f(E)$$

$$\begin{cases} \frac{1}{2}\left\{\dfrac{1}{E'-(E+i\varepsilon)} + \dfrac{1}{E'-(E+i\varepsilon)}\right\} = P\dfrac{1}{E'-E} \\ \frac{1}{2}\left\{\dfrac{1}{E'-(E+i\varepsilon)} - \dfrac{1}{E'-(E-i\varepsilon)}\right\} = \pi i\,\delta(E'-E) \end{cases}$$

$$\begin{cases} \dfrac{1}{E'-E-i\varepsilon} = P\dfrac{1}{E'-E} + i\pi\,\delta(E'-E) \\ \dfrac{1}{E'-E+i\varepsilon} = P\dfrac{1}{E'-E} - i\pi\,\delta(E'-E) \end{cases}$$

$$\boxed{\frac{1}{x\pm i\varepsilon} = P\frac{1}{x} \mp i\pi\,\delta(x)}$$

16.

$$g(z) = \frac{f(z) - \sum \frac{1}{k!} f^{(k)}(E_0)(z-E_0)^k}{(z-E_0)^n}$$

$$|g(z)| \le \frac{|f(z)|\,|z|^n}{|z|^n |z-E_0|^n} + \sum \frac{\frac{1}{k!}|f^{(k)}(E_0)|}{|z-E_0|^{n-k}} \to 0 \quad (z \to \infty)$$

$$\mathrm{Re}\, g(E) = \frac{P}{\pi} \int_{-\infty}^{\infty} \frac{\mathrm{Im}\, g(E')}{E'-E}\, dE'$$

$$\mathrm{Re}\, \hat{f}(E) = \frac{(E-E_0)^n}{\pi} P \int_{-\infty}^{\infty} \frac{dE'\, \mathrm{Im}\, \hat{f}(E')}{(E'-E)(E'-E_0)^n}$$

$n=1$

$$\mathrm{Re}\, f(E) - \mathrm{Re}\, f(E_0) = \frac{E-E_0}{\pi} P \int_{-\infty}^{\infty} \frac{dE'\, \mathrm{Im}\, f(E')}{(E'-E)(E'-E_0)} \quad (\text{1回引き算の式})$$

$$\mathrm{Re}\, f(E) = P \int_{-\infty}^{\infty} \frac{dE'\, \mathrm{Im}\, f(E')}{E'-E}$$

$$\mathrm{Re}\, f(E_0) =$$

17 $|\tilde{z}| = \frac{R_0^2}{r} > R_0$

$$\frac{d\zeta}{\zeta - z} = \frac{\bar{\zeta} - \bar{z}}{|\zeta - z|^2} d\zeta = \frac{R_0 e^{-i\theta} - r e^{-i\varphi}}{R_0^2 - 2R_0 r\cos(\theta-\varphi) + r^2} iR_0 e^{i\theta} d\theta$$

$$\frac{d\zeta}{\zeta - \tilde{z}} = \frac{r^2 - R_0 r e^{i(\theta-\varphi)}}{R_0^2 - 2R_0 r\cos(\theta-\varphi) + r^2} i d\theta$$

$$f(re^{i\varphi}) = \frac{1}{2\pi i}\int \frac{f(\zeta)}{\zeta - z} d\zeta - \frac{1}{2\pi i}\int \frac{f(\zeta)}{\zeta - \tilde{z}} d\zeta$$

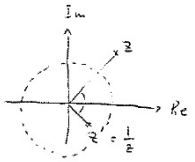
$$= \frac{1}{2\pi}\int_0^{2\pi} \frac{(R_0^2 - r^2) f(R_0 e^{i\theta})}{R_0^2 - 2R_0 r\cos(\theta-\varphi) + r^2} d\theta$$

$$\tilde{z} = \frac{R_0^2}{\bar{z}} = \frac{R_0^2}{r^2} z$$

$$P \quad \frac{r^2}{R_0^2}$$

$$iR_0\left(R_0 - \frac{r^2}{R_0}\right) e^{-i\theta}$$

18. (b) $(z_1, z_2, z_3) \to (0, \infty, 1)$



$$f(z) = \frac{z - z_1}{z - z_2} \Big/ \frac{z_3 - z_1}{z_3 - z_2}$$

$(\omega_1, \omega_2, \omega_3) \to (0, \infty, 1)$

$$g(\omega) = \frac{\omega - \omega_1}{\omega - \omega_2} \Big/ \frac{\omega_3 - \omega_1}{\omega_3 - \omega_2}$$

$(z_1, z_2, z_3) \to (\omega_1, \omega_2, \omega_3)$

$g^{-1} \circ f$

$$\omega = \frac{\alpha z + \beta}{\gamma z + \delta}$$

$$\omega - \frac{\alpha}{\gamma} = \frac{\alpha\delta - \beta\gamma}{\gamma^2} \cdot \frac{1}{z + \frac{\delta}{\gamma}}$$

$$\left|\omega - \frac{\alpha}{\gamma}\right| = \left|\frac{\alpha\delta - \beta\gamma}{\gamma^2}\right| \cdot \frac{1}{\left|z + \frac{\delta}{\gamma}\right|}$$

circle $\to$ circle

$$\omega = \frac{az + b}{cz + d}$$

$a = \epsilon_{ijk} z_i \omega_j$

$b = \epsilon_{ijk} z_i \omega_j z_k \omega_k$

$c = \epsilon_{ijk}\, z_i \omega_j$ (i, j, k = 1, 2, 3) $\leftarrow \sum_k \epsilon_{ijk} z_i \omega_j$

$d = \epsilon_{ijk} z_i z_j \omega_i$

$$\omega = \frac{az + b}{cz + d} = T(z) \qquad T = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

$$\omega = \frac{a'z' + b'}{c'z' + d'} = T'(z')$$

$$z' = \frac{az + b}{cz + d} = T(z)$$

$$\omega = T'(T(z)) = (T' \circ T)(z)$$

19. (a) $\left|k\left(\frac{z-\beta}{z-\alpha}\right)\right| = 1$

垂直二等分線 $|z-\alpha| = |z-\beta|$

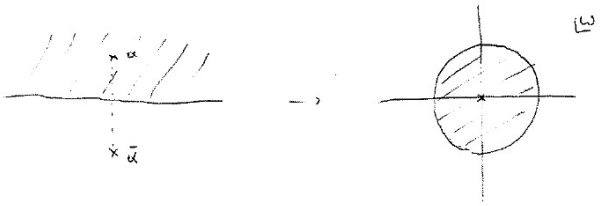
$\rightarrow |k|^2 \frac{|z|^2+|\beta|^2-(\beta\bar{z}+\bar{\beta}z)}{|z|^2+|\alpha|^2-(\alpha\bar{z}+\bar{\alpha}z)} = 1$

実軸 $|z-\alpha| = |z-\bar{\alpha}|$

$|k| \neq 1$ → circle or point or ϕ

$|k| = 1$ → 直線

(b) $\omega = f(z) = e^{i\theta}\frac{z-\alpha}{z-\bar{\alpha}}$ $\quad Im(\alpha) > 0$ $\quad f(\alpha) = 0$



20. $f(\bar{z}) = e^{i\theta}\frac{\bar{z}-\alpha}{\bar{z}-\bar{\alpha}} = \frac{1}{e^{i\theta}\frac{\bar{z}-\bar{\alpha}}{\bar{z}-\alpha}} = \overline{\left(\frac{1}{e^{i\theta}\frac{z-\alpha}{z-\bar{\alpha}}}\right)} = \overline{\left(\frac{1}{\omega}\right)} = \bar{\omega}^{-1} = \tilde{\omega}$

21.

$$\frac{\frac{1}{2}-z}{1} = \frac{x}{\xi} = \frac{y}{\eta}$$

(a) $\xi = \dfrac{x+iy}{\frac{1}{2}-z} = \dfrac{\frac{1}{2}+z}{x-iy}$ $\left(x^2+y^2 = (\tfrac{1}{2}+z)(\tfrac{1}{2}-z)\right)$

(b) $\xi' = e^{i\psi}\xi$ $R_z(\psi) = \begin{pmatrix} e^{i\frac{\psi}{2}} & 0 \\ 0 & e^{-i\frac{\psi}{2}} \end{pmatrix}$

(c) 球面上の回転 → 1次変換 $\left(f(z) = T(z),\ T = \begin{pmatrix} a & b \\ c & d \end{pmatrix}\right)$

(d) $R(\psi,\theta,\varphi) = R_z(\varphi)R_y(\theta)R_z(\psi)$ $R_y(\theta) = \begin{pmatrix} \cos\frac{\theta}{2} & -\sin\frac{\theta}{2} \\ \sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{pmatrix}$

y軸回転 $\begin{pmatrix} z' \\ x' \end{pmatrix} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}\begin{pmatrix} z \\ x \end{pmatrix}$

z軸まわりの ψ 回転

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} \cos\psi & -\sin\psi \\ \sin\psi & \cos\psi \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix}$$

$\rho = x+iy$ $\quad \rho' = e^{i\psi}\rho$

$$\xi = \frac{\rho}{\frac{1}{2}-z}$$

$$\xi' = \frac{\rho'}{\frac{1}{2}-z'} = e^{i\psi}\xi = \frac{e^{i\frac{\psi}{2}}\xi + 0}{0\cdot\xi + e^{-i\frac{\psi}{2}}} \rightarrow \xi' = S(\xi) \quad S = \begin{pmatrix} e^{i\frac{\psi}{2}} & 0 \\ 0 & e^{-i\frac{\psi}{2}} \end{pmatrix}$$

y軸まわりの θ 回転

$$\begin{pmatrix} z'' \\ x'' \end{pmatrix} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}\begin{pmatrix} z' \\ x' \end{pmatrix}$$

$$\rho' = \frac{z'+ix'}{\frac{1}{2}-y'} \quad \text{又は} \quad \rho'' = e^{i\theta}\rho'$$

$$\frac{\rho'+1}{\rho'-1} = \frac{\frac{1}{2}+z'+i(x'+iy')}{-(\frac{1}{2}-z')+i(x'-iy')} \quad \left((\tfrac{1}{2}-z')\xi'\right) = \cdots = \frac{\xi'}{i} \qquad \xi' = i\frac{\rho'+1}{\rho'-1} \rightarrow \rho' = \frac{\xi'+i}{\xi'-i}$$

$$\left(\frac{\frac{1}{2}+z'}{\xi'}\right)$$

$$\xi'' = i\frac{\rho''+1}{\rho''-1} = i\frac{e^{i\theta}\rho'+1}{e^{i\theta}\rho'-1} = i\frac{e^{i\theta}\frac{\xi'+i}{\xi'-i}+1}{e^{i\theta}\frac{\xi'+i}{\xi'-i}-1} = i\frac{(e^{i\theta}+1)\xi'+i(e^{i\theta}-1)}{(e^{i\theta}-1)\xi'+i(e^{i\theta}+1)} = \frac{\cos\frac{\theta}{2}\xi' - \sin\frac{\theta}{2}}{\sin\frac{\theta}{2}\xi' + \cos\frac{\theta}{2}} = S'(\xi')$$

$$S' = \begin{pmatrix} \cos\frac{\theta}{2} & -\sin\frac{\theta}{2} \\ \sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{pmatrix} \qquad S'' = \begin{pmatrix} e^{i\frac{\psi}{2}} & 0 \\ 0 & e^{-i\frac{\psi}{2}} \end{pmatrix}$$

$\xi''' = (S''\circ S'\circ S)\xi$

$\Gamma^{\frac{1}{2}}_{\mu\nu}(\varphi\theta\psi)$

$SU(2)$ の $S=\frac{1}{2}$ 表現 Matrix

$O(3)$ 表示

$$\begin{pmatrix} x''' \\ y''' \\ z''' \end{pmatrix} = \begin{pmatrix} \cos\psi & -\sin\psi & 0 \\ \sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{pmatrix}\begin{pmatrix} \cos\varphi & -\sin\varphi & 0 \\ \sin\varphi & \cos\varphi & 0 \\ 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = R(\varphi\theta\psi)\begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

22

(a) $f_1(z) = \sum_{n=0}^{\infty} z^n = \lim_{n\to\infty} \frac{1-z^n}{1-z} = \frac{1}{1-z} \qquad (|z|<1)$

$f_2(z) = \int_0^{\infty} e^{-(1-z)t}\,dt = \frac{1}{z-1}\left[e^{-(1-z)t}\right]_0^{\infty} = \frac{1}{1-z} \qquad (\mathrm{Re}\, z < 1)$

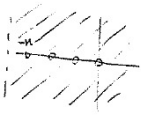
$f_3(z) = -\frac{1}{z}\sum_{n=0}^{\infty}\frac{1}{z^n} = -\frac{1}{z}\cdot\frac{1}{1-\frac{1}{z}} = \frac{1}{1-z} \qquad \left(\left|\frac{1}{z}\right|<1\right)$

(b) $\mathrm{Re}\, z > 0\ ;\ \Gamma(z) = \int_0^{\infty} e^{-t}t^{z-1}\,dt = \left[\frac{1}{z}t^z e^{-t}\right]_0^{\infty} + \frac{1}{z}\int_0^{\infty} e^{-t}t^z\,dt = \frac{1}{z}\Gamma(z+1), \quad \Gamma(1) = 1$

$\downarrow 0$

$-1 < \mathrm{Re}\, z\ ;\ \Gamma(z) = \frac{\Gamma(z+1)}{z} = \frac{1}{z}\int_0^{\infty} e^{-t}t^z\,dt$

$= \frac{1}{z(z+1)}\Gamma(z+2) = \cdots = \frac{1}{z(z+1)\cdots(z+n)}\Gamma(z+n+1) \qquad \mathrm{Re}(z+n+1) > 0$



(C) $\frac{\partial f(z)}{\partial \bar{z}} = 0 \;\to\; \overline{\frac{\partial f(z)}{\partial z}} = 0 \;\to\; \frac{\partial \overline{f(z)}}{\partial z} = 0 \qquad (\mathrm{Im}\, z < 0)$

$f(z) = \frac{1}{2\pi i}\oint_C \frac{f(\zeta)}{\zeta - z}\,d\zeta = \frac{1}{2\pi i}\int_{-\infty}^{\infty}\left(\frac{f(\zeta)-\overline{f(\zeta)}}{\zeta - z}\right)d\zeta = \frac{1}{2\pi i}\int_{-\infty}^{\infty}\frac{f(t)-\overline{f(t)}}{t-z}\,dt = \frac{1}{2\pi i}\int_{-\infty}^{\infty}\frac{2i\,\mathrm{Im}\, f(t)}{t-z}\,dt$

23.

(a) $P(z)$ に零点が n 個 $\rightarrow \oint \frac{P'(z)}{P(z)} dz = 0$

$\downarrow$

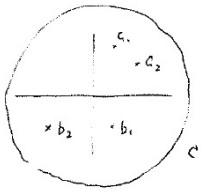
$\int n \frac{1}{z} dz = 2\pi n i$ 積分

(b) $\mathrm{Res}\left(f'/f, a\right) = \begin{cases} n & a \text{ が } f \text{ の } n \text{ 次の零点} \\ -n & \text{〃} \quad \text{極} \end{cases}$

$\frac{1}{2\pi i} \oint \frac{f'(z)}{f(z)} dz = \sum n_i - \sum n_j$

($f(z)$: c_i が n_i 次の零点
b_j が n_j 次の極

例) 有理関数 $f(z)$ の場合 $\rightarrow$ meromorphic
(pole を除いて正則)



$f(z) = \frac{(z-b_1) \cdots (z-b_m)}{(z-a_1) \cdots (z-a_n)}$

$\frac{f'(z)}{f(z)} = \frac{d \log f(z)}{dz} = \frac{d}{dz} \{ \log(z-b_1) + \log(z-b_2) \cdots - \log(z-c_1) - \log(z-c_2) \cdots \}$

$\frac{1}{2\pi i} \oint_C \frac{f'(z)}{f(z)} dz = \frac{1}{2\pi i} \oint_C \frac{d \log f(z)}{dz} dz = \frac{1}{2\pi i} \sum \oint_{C_i} \frac{d \log f(z)}{dz} dz + \frac{1}{2\pi i} \sum \oint$

$= \frac{1}{2\pi i} \sum \oint \frac{dz}{z-b_i} - \frac{1}{2\pi i} \sum \oint_{C_i} \frac{dz}{z-b_j} = \sum - \sum = n - m$

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(a) $f(z) = f(z_0) + \underbrace{f'(z_0)}_{0} r e^{i\varphi} + \frac{1}{2} f''(z_0) r^2 e^{i2\varphi}$

$\underbrace{\qquad\qquad}_{R e^{i\theta}}$

$l_1 : 2\varphi = \pi - \theta$ $\qquad f(z) = f(z_0) + \frac{1}{2} R r^2 e^{i\pi} = f(z_0) - \frac{1}{2} R r^2$

$l_2 : 2\varphi = 2\pi - \theta$ $\qquad = f(z_0) + \frac{1}{2} R r^2 e^{i2\pi} = f(z_0) + \frac{1}{2} R r^2$

(b) $I(\omega) = \int_{C_0} e^{\omega f(z)} g(z) dz \qquad (\omega \gg 1)$

$= \int_{C_0} e^{\omega f(z)} g(z) \frac{dz}{dr} dr$

$= \int_{C_0} e^{\omega (f(z_0) + \frac{1}{2} f''(z_0) r^2)} \cdot \sum_m C_m r^m dr$

(c) $x = \omega z$: $I_1(\omega) = \omega^{\omega+1} \int_0^\infty e^{\omega(\log z - z)} dz \qquad (f(z) = \log z - z)$

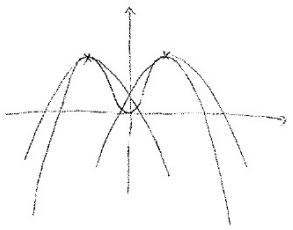
$\approx \omega^{\omega+1} e^{-\omega} \int_0^\infty e^{-\frac{\omega}{2} r^2} dr = \sqrt{2\pi}\, \omega^{\omega+\frac{1}{2}} e^{-\omega}$

$I_2(\omega) = \int_{-\infty}^{\infty} e^{\omega f(z)} dz + \int_0^\infty e^{-\omega r^2} dr = \frac{1}{2}\sqrt{\frac{\pi}{\omega}} \qquad (f(z) = -(z^4 + z^2))$

$I_3(\omega) = e^{\frac{\pi}{4}} \int_{-\infty}^{\infty} e^{-2\omega r^2} dr \times 2 = \sqrt{\frac{\pi}{\omega}} \cdot \sqrt{2}\, e^{\frac{\omega}{4}} \qquad (f(z) = -(z^4 - z^2))$

$\int_{-\infty}^{\infty} e^{\frac{\omega}{4} - 2\omega(x - \frac{1}{\sqrt{2}})^2} dx + \int_{-\infty}^{\infty} e^{\frac{\omega}{4} - 2\omega(x + \frac{1}{\sqrt{2}})^2} dx$

$= 2 e^{\frac{\omega}{4}} \int_0^\infty e^{-2\omega x^2} dx = 2 e^{\frac{\omega}{4}} \left(\frac{\pi}{2\omega}\right)^{\frac{1}{2}}$



$f(z) = \frac{1}{4} + \frac{1}{2}(-4)(x \mp \frac{1}{\sqrt{2}})^2 + \dots$

1. $f(z) = \frac{1}{z^2+a^2}$, $g(z) = \frac{\pi}{\tan \pi z}$

$$\int_C f(z)\,g(z)\,dz = 2\pi i \sum_{n=-m}^{m} \frac{1}{n^2+a^2} + 2\pi i \left[\frac{g(ia)}{2ia} + \frac{g(-ia)}{-2ia} \right]$$

$$= 2\pi i \left[2 \sum_{n=1}^{m} \frac{1}{n^2+a^2} + \frac{1}{a^2} - \frac{\pi}{a} \cdot \coth(\pi a) \right] \xrightarrow[(m\to\infty)]{} 0 \qquad \text{:)} \ |\int| \to 0 \ (m\to\infty)$$

$\left|\frac{1}{z^2+a^2}\right| \cdot \left|\frac{\pi}{\tan \pi z}\right|$ $\int d|z| \sim 2(2m+1) \sim m$

$\lesssim \frac{1}{m^2}$ $< \pi$?

$\tan(ix) = i \tanh x$

$z = x + iy$

$\begin{cases} x = m + \frac{1}{2} \text{ 일 때} & \tan(\pi z) = \tan\left(m\pi + \frac{\pi}{2} + i\pi y\right) \\ -\left(m+\frac{1}{2}\right) \end{cases}$

$$= \tan\left(\frac{\pi}{2} + i\pi y\right) = -\frac{1}{\tan i\pi y} = -\frac{1}{i \tanh \pi y}$$

$$\left|\frac{\pi}{\tan(\pi z)}\right| = |\pi \tanh(\pi y)| \le \pi$$

$y = m + \frac{1}{2}$ $\tan(\pi z) = \tan\left(\pi x + i\left(m\pi + \frac{\pi}{2}\right)\right) \longrightarrow -\frac{1}{i}$ $(m\to\infty)$

$$\sum_{n=1}^{\infty} \frac{1}{n^2+a^2} = \frac{1}{2a} \cdot \frac{\pi}{\tanh(\pi a)} - \frac{1}{2a^2}$$

$$\pi \frac{1 + \frac{(\pi a)^2}{2!} + \cdots}{\pi a + \frac{(\pi a)^3}{3!} + \cdots} = \frac{1}{a} \cdot \left(1 + \underbrace{\frac{(\pi a)^2}{2} - \frac{(\pi a)^2}{6}}_{\frac{(\pi a)^2}{3}} + O(a^4)\right)$$

$$= \frac{1}{2a^2}\left\{1 + \frac{1}{3}(\pi a)^2 + O(a^4) - 1\right\}$$

$$\to \frac{\pi^2}{6} \quad \text{3. 의 증명}$$

2. $f(z) = \dfrac{1}{(z-a)^2}$ $f(z)$ は $z=a$ で 2 位の極

$|g(z)|$ は C_m 上で有界で $|g(z)| < M$ とする.

$$\left| \int_{C_m} f(z) g(z)\, dz \right| \leq \int_{C_m} \frac{M\, dz}{|z-a|^2} \to 0 \quad (m \to \infty)$$

$f(z)\cdot g(z)$ の $z=a$ での留数は $\lim_{z\to a}\left(\frac{d}{dz}\cdot\frac{\pi}{\tan \pi z}\right) = -\frac{\pi^2}{\sin^2 \pi a}$

$$\therefore\ -\frac{\pi^2}{\sin^2 \pi a} + \sum_{n=-\infty}^{\infty} \frac{1}{(n-a)^2} = 0$$

$$\sum_{n=-\infty}^{\infty} \frac{1}{(n-a)^2} = \left(\frac{\pi}{\sin \pi a}\right)^2$$

n 次の pole に対して

$$\operatorname*{Re}_{z=a} f(z) = \lim_{z\to a} \frac{1}{(n-1)!}\left(\frac{\partial}{\partial z}\right)^{n-1} (z-a)^n f(z)$$

$$\operatorname*{Re}_{z=a} \frac{1}{(z-a)^2}\cdot\frac{\pi}{\tan(\pi z)} = \lim_{z\to a} \frac{\partial}{\partial z}\cdot\frac{\pi}{\tan(\pi z)} = -\pi \frac{1}{\tan(\pi z)}\cdot\frac{\pi}{\cos^2(\pi z)}\Big|_{z=a} = \frac{\pi}{\sin^2(\pi a)}$$

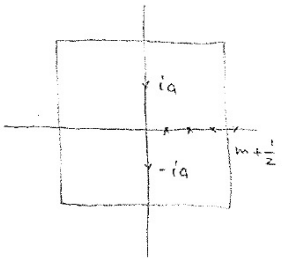
Riemann の ζ - 関数

$$\zeta(z) = \sum_{n=0}^{\infty} \frac{1}{n^z} \quad (\mathrm{Re}\, z > 1)$$

$\zeta(1)$ → 発散

$\zeta(2) = \dfrac{\pi^2}{6}$ ⌋ 3. の答

4. $$I = \frac{1}{2\pi i}\oint \frac{e^{iz\xi}}{z^2+a^2}\cdot\frac{2\pi i}{e^{2\pi i z}-1}\,dz$$



留数 $z = m \in \mathbb{Z}$ $\quad \frac{e^{in\xi}}{n^2+a^2}$

$z = ia$

$$\frac{e^{-a\xi}}{2ia}\cdot\frac{2\pi i}{e^{-2\pi a}-1} = \frac{\pi}{a}\cdot\frac{e^{-a\xi}}{e^{-2\pi a}-1}$$

$z = -ia$

$$\frac{e^{a\xi}}{-2ia}\cdot\frac{2\pi i}{e^{2\pi a}-1} = -\frac{\pi}{a}\cdot\frac{e^{a\xi}}{e^{2\pi a}-1}$$

$$I = \sum_{n=-\infty}^{\infty}\frac{e^{in\xi}}{n^2+a^2} + \frac{\pi}{a^2}\left(\frac{e^{-a\xi}}{e^{-2\pi a}-1} - \frac{e^{a\xi}}{e^{2\pi a}-1}\right)$$

$$= -\frac{\cosh a(\xi-\pi)}{\sinh(\pi a)}$$

$I = 0$ より $\quad \sum_{n=-\infty}^{\infty}\frac{e^{in\xi}}{n^2+a^2} = \frac{1}{a^2} + \sum_{n=1}^{\infty}\frac{e^{in\xi}+e^{-in\xi}}{n^2+a^2}$

$$= \frac{1}{a^2} + 2\sum_{n=1}^{\infty}\frac{\cos n\xi}{n^2+a^2}$$

$$= \frac{\pi}{a}\cdot\frac{\cosh a(\xi-\pi)}{\sinh(\pi a)} \quad\to\quad \sum_{n=1}^{\infty}\frac{\cos n\xi}{n^2+a^2} = \frac{\pi}{2a}\frac{\cosh a(\xi-\pi)}{\sinh(\pi a)} - \frac{1}{2a^2}$$

$$f = \left|\frac{e^{iz\xi}}{e^{2\pi i z}-1}\right| = \left|\frac{e^{i\xi x - \xi y}}{e^{2\pi i x - 2\pi y}-1}\right| = \frac{e^{-\xi y}}{\left|e^{-2\pi y}(\cos 2\pi x + i\sin 2\pi y) - 1\right|} = \frac{e^{-\xi y}}{\sqrt{e^{-4\pi y}+1-2e^{-2\pi y}\cos(2\pi x)}}$$

$x = \pm\left(m+\frac{1}{2}\right)$ $\qquad f \le \frac{e^{-\xi y}}{1+e^{-2\pi y}} = \begin{cases} e^{-\xi y} \le 1 & \text{for } y \ge 0 \quad (0 \le \xi < 2\pi) \\ e^{(2\pi-\xi)y} \le 1 & \text{for } y \le 0 \end{cases}$

$y = m+\frac{1}{2}$ $\quad f \le \frac{e^{-\xi(m+\frac{1}{2})}}{1-e^{-2\pi(m+\frac{1}{2})}} \xrightarrow[(m\to\infty)]{} \le 1$

$y = -m-\frac{1}{2}$ $\quad f \le \frac{e^{\xi(m+\frac{1}{2})}}{e^{2\pi(m+\frac{1}{2})}-1} \xrightarrow[(m\to\infty)]{} e^{(\xi-2\pi)(m+\frac{1}{2})} \le 1$

5.

$$\oint \frac{e^{-z^2\pi t}}{e^{2\pi i z}-1}\,dz \xrightarrow[N\to\infty]{} \sum_{n=-\infty}^{\infty} \operatorname{Re}\frac{2\pi i}{e^{2\pi i z}-1}\, e^{-z^2\pi t} = \sum_{n=-\infty}^{\infty} e^{-n^2\pi t} = f(t)$$

$$\frac{2\pi i}{e^{2\pi i z}-1} \sim \frac{1}{z}$$

$$\left|\int_{L_4}\right| \le 2\left|\frac{e^{-(N+\frac12+iy)^2\pi t}}{e^{2\pi i(N+\frac12)-2\pi y}-1}\right| = \frac{e^{-(N+\frac12)^2\pi t}\cdot e^{y^2\pi t}}{e^{-2\pi y}+1} \propto e^{-(N+\frac12)^2\pi t} \longrightarrow 0 \quad (N\to\infty)$$

$$\frac{1}{e^{2\pi i z}-1} = -\frac{1}{1-e^{2\pi i z}} = -\sum_{n=0}^{\infty}\left(e^{2\pi i z}\right)^n = -\sum_{n=0}^{\infty} e^{2\pi i n z}$$

$$e^{2\pi i z} = \begin{cases} e^{2\pi i x}\, e^{-2\pi} & z = x+i \text{ on } L_1 \\ e^{2\pi i x}\, e^{2\pi} & z = x-i \text{ on } L_3 \end{cases}$$

L_3 에서

$$\frac{1}{e^{2\pi i z}-1} = \frac{1}{e^{2\pi i z}}\cdot\frac{1}{1-e^{-2\pi i z}} = e^{-2\pi i z}\sum_{n=0}^{\infty} e^{-2\pi i n z} = \sum_{n=0}^{\infty} e^{-2\pi i(n+1)z} = \sum_{n=1}^{\infty} e^{-2\pi i n z}$$

$$\int_{L_1} = -\sum_{n=0}^{\infty}\int_{L_1} e^{-z^2\pi t + 2\pi i n z}\,dz \qquad z = x+i$$

$$\int_{L_3} = \sum_{n=1}^{\infty}\int_{L_3} e^{-z^2\pi t - 2\pi i n z}\,dz$$

$$\exp \text{의 꼴} = -(x+i)^2\pi t + 2\pi i n(x+i)$$

$$= -\pi t\left(x - i\frac{-t+n}{t}\right)^2 - \frac{\pi(-t+n)^2}{t} + \pi t - 2\pi n$$

$$= -\pi t\left(x - \frac{-t+n}{t}\right)^2 - \frac{\pi n^2}{t}$$

$$\int_{L_1} = \sum_{n=0}^{\infty}\int_{-\infty}^{\infty} dx\, e^{-\pi t\left(x - i\frac{t+n}{t}\right)^2} e^{\frac{\pi n^2}{t}} = \frac{1}{\sqrt{t}}\sum_{n=0}^{\infty} e^{\frac{\pi n^2}{t}}$$

$$\int_{L_2} = \frac{1}{\sqrt{t}}\sum_{n=1}^{\infty} e^{\frac{\pi n^2}{t}}$$

$$f(t) = \frac{1}{\sqrt{t}}\sum_{n=-\infty}^{\infty} e^{\frac{-\pi n^2}{t}} = \frac{1}{\sqrt{t}}\, f\left(\frac{1}{t}\right)$$

5. Poisson の和公式

のつづき

$$\vartheta(x) = \sum_{n=-\infty}^{\infty} e^{-\pi n^2 x} \quad \text{テータ関数} \quad (x>0)$$

$$= \frac{1}{\sqrt{x}} \sum_{n=-\infty}^{\infty} e^{-\pi \frac{n^2}{x}} = \frac{1}{\sqrt{x}} \vartheta\left(\frac{1}{x}\right)$$

$$f(\tau) = e^{-\frac{x}{4\pi}\tau^2}$$

$$f(2\pi n) = e^{-\pi n^2 x}$$

$$\int_{-\infty}^{\infty} f(\tau) e^{-in\tau} d\tau = \int_{-\infty}^{\infty} e^{-\frac{x}{4\pi}\tau^2 - in\tau} d\tau = \int_{-\infty}^{\infty} e^{-\frac{x}{4\pi}\left(\tau + i\frac{2\pi}{x}n\right)^2} \cdot e^{-\frac{\pi n^2}{x}} d\tau$$

$$= \left(\frac{4\pi^2}{x}\right)^{\frac{1}{2}} e^{-\frac{\pi n^2}{x}} = 2\pi \frac{1}{\sqrt{x}} e^{-\frac{\pi n^2}{x}}$$

$$\underbrace{\frac{1}{2\pi}\int_{-\infty}^{\infty} f(\tau) e^{-in\tau} d\tau} = \frac{1}{\sqrt{x}} \sum_{n=-\infty}^{\infty} e^{-\frac{\pi n^2}{x}} = \sum_{n=-\infty}^{\infty} e^{-\pi n^2 x} = \underbrace{\sum_{n=-\infty}^{\infty} f(2\pi n)} \quad \text{Poisson の和公式}$$

$$\sum_{m=-\infty}^{\infty} f(2\pi m) = \frac{1}{2\pi} \sum_{n=-\infty}^{\infty} \int_{-\infty}^{\infty} f(\tau) e^{-in\tau} d\tau \quad (\text{Poisson の和公式})$$

ここに $\sum_{m=-\infty}^{\infty} f(2\pi m + \tau)$ が $0 \le \tau < 2\pi$ において一様収束し，かつ Fourier 級数に展開できるとする．

(証明)

$F(\theta) \equiv \sum_{m=-\infty}^{\infty} f(2\pi m + \theta)$ 一様収束で Fourier 級数展開可能（仮定より）

$F(\theta) = \sum_{n=-\infty}^{\infty} F_n e^{in\theta}$ とすると $F_n = \frac{1}{2\pi}\int_0^{2\pi} e^{-in\theta} F(\theta) d\theta$

$$F(\theta) = \frac{1}{2\pi}\int_0^{2\pi} e^{-in\theta} \sum_{m=-\infty}^{\infty} f(2\pi m + \theta) d\theta$$

$$= \frac{1}{2\pi} \sum_{m=-\infty}^{\infty} \int_0^{2\pi} e^{-in\theta} f(2\pi m + \theta) d\theta \quad \leftarrow \text{一様収束}$$

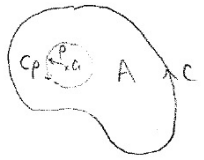
$$= \frac{1}{2\pi} \sum_{m=-\infty}^{\infty} \int_{2\pi m}^{2\pi(m+1)} e^{-in\theta} f(\theta) d\theta$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-in\theta} f(\theta) d\theta$$

$F(0) = \sum_{n=-\infty}^{\infty} F_n$ で QED

6.

$$F(z) = \frac{f(z) - f(a)}{g(z) - g(a)}\, g'(z)$$



$F(z)$ は A 内で analytic ($z=a$ で $F(a) = f'(a)$)

Cauchy の定理より $\int_C F(z)dz = 0$ $\quad g'(z) \neq 0$ (1次の pole)

$$\therefore \quad f(a)\int_C \frac{g'(z)}{g(z) - g(a)}\, dz = \int_C \frac{f(z)}{g(z) - g(a)}\, g'(z)\, dz$$

$g(z) - g(a) = (z-a)\, g_1(z)$ $\quad (g_1(a) \neq 0)$ とおくと

$$\int_C \frac{g'(z)}{g(z) - g(a)}\, dz = \int_C \frac{g_1(z) + (z-a)\, g_1'(z)}{(z-a)\, g_1(z)}\, dz = \int_C \frac{dz}{z-a} + \int_C \frac{g_1'}{g_1}\, dz$$

$$\therefore f(a) = \frac{1}{2\pi i}\int_C \frac{f(z)}{g(z) - g(a)}\, g'(z)\, dz$$

$f(z) = z$ とおくと

$$z = \frac{1}{2\pi i}\int_C \frac{\zeta\, g'(\zeta)}{g(\zeta) - g(z)}\, d\zeta$$

$t = g(z)$ とし $z - z_0 = \sum_{n=1}^{\infty} a_n t^n$ とおく $\quad g(z_0) = 0$

$$a_n = \frac{1}{2\pi i}\int_C \frac{g^{-1}(t)}{t^{n+1}}\, dt = \frac{1}{2\pi i}\int_C \frac{\zeta\, g'(\zeta)}{g(\zeta)^{n+1}}\, d\zeta$$

$t = g(z)$, $0 = g(z_0)$

$$\frac{\zeta\, g'(\zeta)}{g(\zeta) - g(z)} = \frac{\zeta\, g'(\zeta)}{g(\zeta)} \cdot \frac{1}{1 - \dfrac{g(z)}{g(\zeta)}} = \frac{\zeta\, g'(\zeta)}{g(\zeta)} \sum_{n=0}^{\infty} \left(\frac{g(z)}{g(\zeta)}\right)^n = \sum_{n=0}^{\infty} \frac{\zeta\, g'(\zeta)}{g(\zeta)^{n+1}} (g(z))^n$$

一様収束から

$$z = \sum_{n=0}^{\infty} \left(\frac{1}{2\pi i}\int_C \frac{\zeta\, g'(\zeta)}{g(\zeta)^{n+1}}\, d\zeta\right) \cdot (g(z))^n = \sum_{n=0}^{\infty} a_n t^n$$

$\left|\dfrac{g(z)}{g(\zeta)}\right| < r_0 < 1$ $\quad z \in d$ の内部

$\zeta \in C$

7. $x = 1 + y\,x^m$, $y = \dfrac{x-1}{x^m}$ $\longleftrightarrow$ $t = g(z)$ $\qquad x = e^{\ln x}$

$t \to y$, $z \to x$

$g(z) \to \dfrac{x-1}{x^m}$

$\dfrac{x-1}{x^m} = 0 \to x = 1 = z_0$

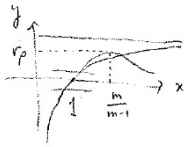
$f(z) = \log z$

$f(z) - f(z_0) = \sum_{n=1}^{\infty} a_n t^n$ $\qquad a_n = \dfrac{1}{2\pi i}\oint_c \dfrac{\log\zeta\, g'(\zeta)}{g(\zeta)^{n+1}}\,d\zeta$

$\log x = \sum_{n=1}^{\infty} a_n y^n$ $\qquad a_n = \dfrac{1}{2\pi i}\oint \dfrac{\log\zeta \left(\frac{\zeta-1}{\zeta^m}\right)'}{\left(\frac{\zeta-1}{\zeta^m}\right)^{n+1}}\,d\zeta$

$m = 1$

$y = \dfrac{x-1}{x} = g(x)$



$r_p = g\left(\frac{m}{m-1}\right) = m^{-m}(m-1)^{m-1}$

$= \dfrac{1}{2\pi i}\oint \dfrac{\log\zeta}{g(\zeta)^n}\cdot\dfrac{g'(\zeta)}{g(\zeta)}\,d\zeta$

$= \dfrac{1}{2\pi i}\oint \dfrac{\log\zeta}{g(\zeta)^n}\cdot\dfrac{d}{d\zeta}\log(g(\zeta))\,d\zeta$

$= \dfrac{1}{2\pi i}\oint \log\zeta\left(\dfrac{\zeta^m}{\zeta-1}\right)^n\left(\dfrac{1}{\zeta-1} - \dfrac{m}{\zeta}\right)d\zeta$

$= \dfrac{1}{2\pi i}\oint (\log\zeta)\cdot\dfrac{\zeta^{nm-1}}{(\zeta-1)^{n+1}}\{(1-m)\zeta + m\}\,d\zeta$

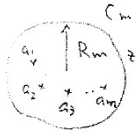
$= \operatorname{Res}_{z=1}\{\cdots\}$

$= \lim_{\zeta\to 1}\dfrac{1}{n!}\left(\dfrac{\partial}{\partial\zeta}\right)^n(\zeta-1)^{n+1}\dfrac{\zeta^{nm-1}}{(\zeta-1)^{n+1}}\{(1-m)\zeta + m\}\log\zeta$

$= \dfrac{1}{n!}\left(\dfrac{\partial}{\partial\zeta}\right)^n\{(1-m)\zeta^{nm} + m\zeta^{nm}\}\log\zeta\Big|_{\zeta=1}$

8.

$f(z)$: C_m 上で 有界 $(m = 0, 1, 2, \cdots)$



$$\frac{1}{2\pi i}\oint_{C_m}\frac{f(z)}{z-x}\,dz = f(x) + \sum_{m=1}^{N}\frac{1}{a_m - x}\operatorname*{Res}_{z=a_m} f(z)$$

C_m として 原点 $z=0$ からの距離 R_m が $m\to\infty$ で $R_m\to\infty$ となるものとする.

$\frac{1}{z-x} = \frac{1}{z} + \frac{x}{z(z-x)}$ と使うと

$$\frac{1}{2\pi i}\oint_{C_m}\frac{f(z)}{z-x}\,dz = \frac{1}{2\pi i}\oint_{C_m}\frac{f(z)}{z}\,dz + \frac{x}{2\pi i}\oint_{C_m}\frac{f(z)}{z(z-x)}\,dz$$

$$= f(0) + \sum_{m=1}^{N_m}\operatorname*{Res}_{z=a_m} f(z) + \underbrace{\frac{x}{2\pi i}\oint\frac{f(z)}{z(z-x)}\,dz}_{R_m\to\infty} \le \int_0^{2\pi}\frac{|f(z)|}{|z|}\,d\theta \to 0$$

$z = z''$

$$f(x) = f(0) + \sum_{m=1}^{\infty}\left\{\frac{1}{x-a_m} + \frac{1}{a_m}\right\}\operatorname*{Res}_{z=a_m} f(z)$$

9. δ で $f(z) = \frac{1}{\sin z} - \frac{1}{z}$ とおく $f(0) = 0$, $f(z)$ の pole は $z = n\pi$ $(n\neq 0)$ である.

$$\operatorname*{Res}_{z=n\pi}[f(z)] = \frac{1}{\cos n\pi} = (-1)^n \qquad \therefore \quad \frac{1}{\sin z} = \frac{1}{z} + \sum_{\substack{n=-\infty\\ n\neq 0}}^{\infty}(-1)^n\left\{\frac{1}{z-n\pi} + \frac{1}{n\pi}\right\}$$

$$\lim_{N\to\infty}\sum_{n=1}^{N}\left(\frac{1}{z-n\pi} + \frac{1}{z+n\pi} + \frac{1}{n\pi} - \frac{1}{n\pi}\right)(-1)^n = \lim_{N\to\infty}\sum_{n=1}^{N}\frac{2z}{z^2-n^2\pi^2}(-1)^n$$

10.

$f(z) \to \dfrac{f'(z)}{f(z)}$ とおきかえると $\left|\dfrac{f'(z)}{f(z)}\right| < M$ （C_n上で有界）

$$\frac{f'(z)}{f(z)} = \frac{f'(0)}{f(0)} + \sum \left\{ \frac{1}{z-a_n} + \frac{1}{a_n} \right\} \operatorname*{Res}_{z=a_n} \left(\frac{f'(z)}{f(z)} \right)$$

$$\operatorname*{Res}_{z=a_n} \frac{f'(z)}{f(z)} = \lim_{z \to a_n} (z-a_n) \frac{f'(z)}{f(z)-f(a_n)} \overset{\frac{0}{0}}{=} \frac{f'(a_n)}{f'(a_n)} = 1 \quad \text{対}$$

$$\frac{f'(z)}{f(z)} = \frac{f'(0)}{f(0)} + \sum \left\{ \frac{1}{z-a_n} + \frac{1}{a_n} \right\} \quad \text{これぞれ積分すると}$$

$$\log f(z) - \log f(0) = \frac{f'(0)}{f(0)} z + \sum \left(\log (z-a_n) + \frac{z}{a_n} \right) - \sum \log (-a_n)$$

$$= \frac{f'(0)}{f(0)} z + \sum \left\{ \log \left(1 - \frac{z}{a_n}\right) + \frac{z}{a_n} \right\}$$

？ 一様収束すれば

$$f(z) = f(0)\, e^{\frac{f'(0)}{f(0)} z}\, e^{\sum \log \left(1-\frac{z}{a_n}\right) + \frac{z}{a_n}}$$

$$= f(0)\, e^{\frac{f'(0)}{f(0)} z} \prod \left\{ \left(1 - \frac{z}{a_n}\right) e^{z/a_n} \right\}$$

$f(z) = \sum_{n=1}^{\infty} \left\{ \dfrac{1}{x-a_n} + \dfrac{1}{a_n} \right\}$ で $\sum_{n=1}^{\infty} \left(\dfrac{1}{a_n}\right)^2$ が収束すれば絶対一様収束

(Proof) $\displaystyle\sum_{n=N}^{\infty} \left| \frac{1}{x-a_n} + \frac{1}{a_n} \right| = \sum_{n=N}^{\infty} \left| \frac{x}{a_n (x-a_n)} \right| = |x| \sum_{n=N}^{\infty} \frac{1}{|a_n|^2 \left|1 - \frac{x}{a_n}\right|} \lesssim a_n \sum_{n=N}^{\infty} \frac{1}{|a_n|^2}$

$\left|1 - \frac{x}{a_n}\right| \to 1$

無限積の収束

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$P = (1+u_1)(1+u_2)\cdots$

は $1 + (u_1) + (u_2) + \cdots$ が収束すれば絶対一様収束

$$\left| \left(1 - \frac{z}{a_n}\right) e^{\frac{z}{a_n}} - 1 \right| \sim \left| \frac{z}{a_n} \right|^2$$